

Esercizi

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1. *Risolvere le seguenti equazioni lineari incomplete:*

(a) $\sqrt{3}\sin x - \cos x = 0$

(c) $2\sin x = \sin x + \cos(-x)$

(b) $\sin x - \cos x = 0$

(d) $3(\sin x + \cos x) = \sqrt{3}x(1 + \sqrt{3})$

2. *Risolvere le seguenti equazioni lineari complete con il metodo grafico:*

(a) $\sin x - \cos x = -1$

(d) $1 + 2\cos x + \sin x = 0$

(b) $\sin x + \cos x = 1$

(e) $4\sin x + 4\cos x - 4 = 0$

(c) $3 + \cos x - 3\sin x = 0$

(f) $\sqrt{3}\sin x + 5\cos x + 4 = 0$

3. *Risolvere le seguenti equazioni omogenee:*

(a) $\sin^2 x - \sin x \cos x = 0$

(e) $\sin^x - 2\sin x \cos x + \cos^2 x = 0$

(b) $\sin x \cos x - \sqrt{3}\sin^2 x = 0$

(f) $\sin^2 x - \sin x \cos x - 2\cos^2 x = 0$

(c) $\cos^2 x - \sqrt{3}\sin x \cos x = 0$

(g) $\sin^2 x + 3\cos^2 x + 2\sqrt{3}\sin x \cos x = 0$

(d) $\cos^2 x - 2\sin x \cos x = 0$

4. *Risolvere le seguenti equazioni del tipo $a\sin^2 x + b\sin x \cos x + c\cos^2 x + d = 0$:*

(a) $\sqrt{3}\sin x + \cos x = 1 - \sin^x$

(d) $1 + 2\cos^2 x - 2\sqrt{3}\sin x \cos x = 0$

(b) $2\sin^2 x + \sqrt{3}\sin x \cos x + \cos^2 x = 2$

(e) $1 - 2\sin x \cos x - \cos^2 x = 0$

(c) $1 - 2\sin x \cos x = 4\cos^2 x$

(f) $(1 - \sqrt{3})\sin^x - (1 + \sqrt{3})\sin x \cos x = -\sqrt{3}$